

MENENTUKAN TAUTOLOGI DENGAN 3 PROPOSISI

Kelompok 6

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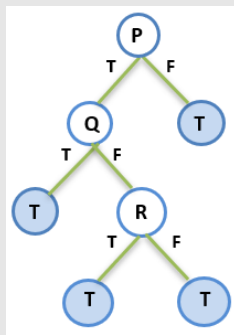
Question 1.

Tentukan apakah 3 proposisi ini tautologi atau kontradiksi dengan metode semantic tree dan falsification method. $(P \rightarrow (Q \vee R)) \rightarrow ((P \wedge (\neg Q)) \rightarrow R)$

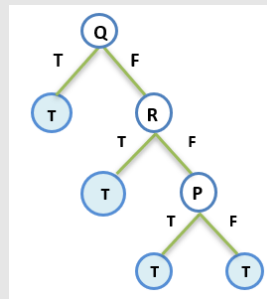
Solutions :

(A) SEMANTIC TREE

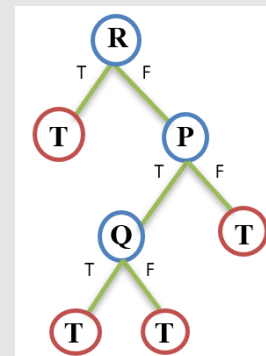
(i) Kepala P :



(ii) Kepala Q :



(iii) Kepala R :



(i) Kepala P

(a) $P = T$

$$(T \rightarrow (Q \vee R)) \rightarrow ((T \wedge (\neg Q)) \rightarrow R)$$

$$(T \rightarrow (?)) \rightarrow ((?) \rightarrow R)$$

*) $Q = T$

$$(T \rightarrow (T \vee R)) \rightarrow ((T \wedge (F)) \rightarrow R)$$

$$(T \rightarrow (T)) \rightarrow ((F) \rightarrow R)$$

$$(T \rightarrow T) = \mathbf{T}$$

*) $Q = F$

$$(T \rightarrow (F \vee R)) \rightarrow ((T \wedge (T)) \rightarrow R)$$

$$(T \rightarrow (?)) \rightarrow ((T) \rightarrow R)$$

**) $R = T$

$$(T \rightarrow (F \vee T)) \rightarrow ((T \wedge (T)) \rightarrow T)$$

$$(T \rightarrow (T)) \rightarrow (T \rightarrow T)$$

$$(T \rightarrow T) = \mathbf{T}$$

**) $R = F$

$$(T \rightarrow (F \vee F)) \rightarrow ((T \wedge (T)) \rightarrow F)$$

$$(T \rightarrow (F)) \rightarrow (T \rightarrow F)$$

$$(F \rightarrow F) = \mathbf{T}$$

(b) $P = F$

$$(F \rightarrow (Q \vee R)) \rightarrow ((F \wedge (\neg Q)) \rightarrow R)$$

$$(F \rightarrow (?)) \rightarrow (F \rightarrow R)$$

$$(T \rightarrow T) = \mathbf{T}$$

(ii) Kepala Q

(a) **$Q = T$**

$$(P \rightarrow (T \vee R)) \rightarrow ((P \wedge (F)) \rightarrow R)$$

$$(P \rightarrow T) \rightarrow (F \rightarrow R)$$

$$(T \rightarrow T) = \mathbf{T}$$

(b) **$Q = F$**

$$(P \rightarrow (F \vee R)) \rightarrow ((P \wedge (T)) \rightarrow R)$$

$$(P \rightarrow (?)) \rightarrow ((?) \rightarrow R)$$

***) $R = T$**

$$(P \rightarrow (F \vee T)) \rightarrow ((P \wedge (T)) \rightarrow T)$$

$$(P \rightarrow (T)) \rightarrow ((?) \rightarrow T)$$

$$(T \rightarrow T) = \mathbf{T}$$

***) $R = F$**

$$(P \rightarrow (F \vee F)) \rightarrow ((P \wedge (T)) \rightarrow F)$$

$$(P \rightarrow (F)) \rightarrow ((?) \rightarrow F)$$

****) $P = T$**

$$(T \rightarrow (F \vee F)) \rightarrow ((T \wedge (T)) \rightarrow F)$$

$$(T \rightarrow (F)) \rightarrow (T \rightarrow F)$$

$$(F \rightarrow F) = \mathbf{T}$$

****) $P = F$**

$$(F \rightarrow (F \vee F)) \rightarrow ((F \wedge F) \rightarrow F)$$

$$(F \rightarrow (F)) \rightarrow (F \rightarrow F)$$

$$(T \rightarrow T) = \mathbf{T}$$

(iii) Kepala R

(a) **$R = T$**

$$(P \rightarrow (Q \vee T)) \rightarrow ((P \wedge (\neg Q)) \rightarrow T)$$

$$(P \rightarrow T) \rightarrow (T)$$

$$(T \rightarrow T) = \mathbf{T}$$

(b) **$R = F$**

$$(P \rightarrow (Q \vee F)) \rightarrow ((P \wedge (\neg Q)) \rightarrow F)$$

$$(P \rightarrow (?)) \rightarrow ((?) \rightarrow F)$$

***) $P = F$**

$$(F \rightarrow (Q \vee F)) \rightarrow ((F \wedge (\neg Q)) \rightarrow F)$$

$$(T) \rightarrow (F \rightarrow F)$$

$$(T \rightarrow T) = \mathbf{T}$$

***) $P = T$**

$$(T \rightarrow (Q \vee F)) \rightarrow ((T \wedge (\neg Q)) \rightarrow F)$$

$$((?) \rightarrow (?))$$

****) $Q = T$**

$$(T \rightarrow (T \vee F)) \rightarrow ((T \wedge F) \rightarrow F)$$

$$(T \rightarrow T) \rightarrow (F \rightarrow F)$$

$$(T \rightarrow T) = \mathbf{T}$$

****) $Q = F$**

$$(T \rightarrow (F \vee F)) \rightarrow ((T \wedge T) \rightarrow F)$$

$$(T \rightarrow F) \rightarrow (T \rightarrow F)$$

$$(F \rightarrow F) = \mathbf{T}$$

(B) FALSIFICATION METHOD

$$(i) (P \rightarrow (Q \vee R)) \rightarrow ((P \wedge (\neg Q)) \rightarrow R) \equiv F \dots (1)$$

$$(ii) (P \rightarrow (Q \vee R)) \equiv T \dots (2)$$

$$(iii) ((P \wedge (\neg Q)) \rightarrow R) \equiv F \dots (3)$$

Dari 3

$$(iv) ((P \wedge (\neg Q)) \rightarrow R) \equiv F$$

$$(P \wedge (\neg Q)) \equiv T \dots (4)$$

$$R \equiv F \dots (5)$$

Dari 4

$$(v) P \wedge (\neg Q) \equiv T$$

$$P \equiv T \dots (6)$$

$$\neg Q \equiv T \rightarrow Q \equiv F \dots (7)$$

(vi) Substitusi (5), (6), (7), ke persamaan (2)

$$(P \rightarrow (Q \vee R)) \equiv T$$

$$(T \rightarrow (F \vee F)) \equiv T$$

$$(T \rightarrow F) \equiv T$$

$$F \equiv T \text{ (Kontradiksi)} \rightarrow T \text{ AUTOLOGI}$$

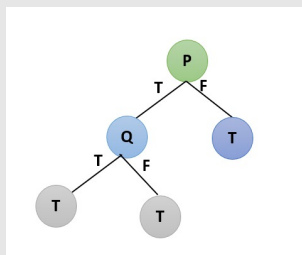
Question 2.

Tentukan apakah 3 proposisi ini tautologi atau kontradiksi dengan metode semantic tree dan falsification method. $(P \rightarrow (Q \wedge R)) \rightarrow ((P \rightarrow Q) \vee (P \rightarrow R))$

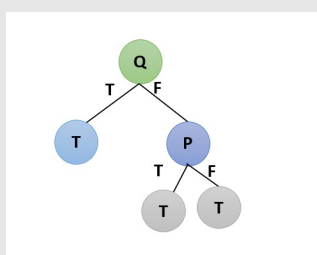
Solutions :

(A) SEMANTIC TREE

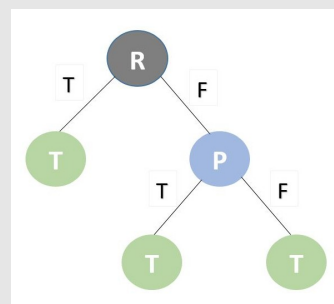
(i) Kepala P :



(ii) Kepala Q :



(iii) Kepala R :



(i) Kepala P

(a) $P = F$

$$(F \rightarrow (Q \wedge R)) \rightarrow ((F \rightarrow Q) \vee (F \rightarrow R))$$

$$(F \rightarrow (?)) \rightarrow (T \vee T)$$

$$(T \rightarrow T) = \mathbf{T}$$

(b) $P = T$

$$(T \rightarrow (Q \wedge R)) \rightarrow ((T \rightarrow Q) \vee (T \rightarrow R))$$

$$(T \rightarrow (?)) \rightarrow ((T \rightarrow ?) \vee (T \rightarrow ?))$$

*) $Q = F$

$$(T \rightarrow (F \wedge R)) \rightarrow ((T \rightarrow F) \vee (T \rightarrow R))$$

$$(T \rightarrow F) \rightarrow (F \vee (?))$$

$$(F \rightarrow ?) = \mathbf{T}$$

*) $Q = T$

$$(T \rightarrow (T \wedge R)) \rightarrow ((T \rightarrow T) \vee (T \rightarrow R))$$

$$(T \rightarrow (?)) \rightarrow (T \vee (?))$$

$$((?) \rightarrow T) = \mathbf{T}$$

(ii) Kepala Q

(a) $Q = T$

$$(P \rightarrow (T \wedge R)) \rightarrow ((P \rightarrow T) \vee (P \rightarrow R))$$

$$(P \rightarrow (?)) \rightarrow (T \vee (?))$$

$$(? \rightarrow T) = \mathbf{T}$$

(b) $Q = F$

$$(P \rightarrow (F \wedge R)) \rightarrow ((P \rightarrow F) \vee (P \rightarrow R))$$

$$(P \rightarrow F) \rightarrow ((?) \vee (?))$$

*) $P = T$

$$(T \rightarrow (F \wedge R)) \rightarrow ((T \rightarrow F) \vee (T \rightarrow R))$$

$$(T \rightarrow F) \rightarrow (F \vee (?))$$

$$(F \rightarrow (?)) = \mathbf{T}$$

*) $P = F$

$$(F \rightarrow (F \wedge R)) \rightarrow ((F \rightarrow F) \vee (F \rightarrow R))$$

$$(F \rightarrow F) \rightarrow (T \vee T)$$

$$(T \rightarrow T) = \mathbf{T}$$

(iii) Kepala R

(a) $R = T$

$$(P \rightarrow (Q \wedge T)) \rightarrow ((P \rightarrow Q) \vee (P \rightarrow T))$$

$$(P \rightarrow (?)) \rightarrow ((?) \vee T)$$

$$((?) \rightarrow T) = \mathbf{T}$$

(b) $R = F$

$$(P \rightarrow (Q \wedge F)) \rightarrow ((P \rightarrow Q) \vee (P \rightarrow F))$$

$$(P \rightarrow F) \rightarrow ((?) \vee (?))$$

*) $P = T$

$$(T \rightarrow (Q \wedge F)) \rightarrow ((T \rightarrow Q) \vee (T \rightarrow F))$$

$$(T \rightarrow F) \rightarrow ((?) \vee F)$$

$$(F \rightarrow (?)) = \mathbf{T}$$

*) $P = F$

$$(F \rightarrow (Q \wedge F)) \rightarrow ((F \rightarrow Q) \vee (F \rightarrow F))$$

$$(F \rightarrow F) \rightarrow (T \vee T)$$

$$(T \rightarrow T) = \mathbf{T}$$

(B) FALSIFICATION METHOD

$$(i) (P \rightarrow (Q \wedge R)) \rightarrow ((P \rightarrow Q) \vee (P \rightarrow R)) \equiv F \dots (1)$$

$$(ii) (P \rightarrow (Q \wedge R)) \equiv T \dots (2)$$

$$(iii) ((P \rightarrow Q) \vee (P \rightarrow R)) \equiv F \dots (3)$$

Dari 3

$$(iv) ((P \rightarrow Q) \vee (P \rightarrow R)) \equiv F$$

$$(P \rightarrow Q) \equiv F \dots (4)$$

$$(P \rightarrow R) \equiv F \dots (5)$$

Dari 4

$$(v) (P \rightarrow Q) \equiv F$$

$$P \equiv T \dots (6)$$

$$Q \equiv F \dots (7)$$

$$(vi) \text{ Substitusi (5), (6), (7), ke persamaan (2)}$$

$$(P \rightarrow (Q \wedge R)) \equiv T$$

$$(T \rightarrow (F \wedge R)) \equiv T$$

$$(T \rightarrow F) \equiv T$$

$$F \equiv T \text{ (Kontradiksi)} \rightarrow T \text{ AUTOLOGI}$$

Question 3.

Premis 1 : Jika saya pintar, maka nilai saya bagus.

Premis 2 : Jika nilai saya bagus, maka saya cepat lulus.

Apakah dapat diambil kesimpulan dari kedua premis di atas? Coba dengan cara operator dasar dan cara tabel kebenaran!

Solutions :

P = Saya pintar

Q = Nilai saya bagus

R = Saya cepat lulus

(A) OPERATOR DASAR

Jika saya pintar, maka nilai saya akan bagus. $P \rightarrow Q$

Jika nilai saya bagus, maka saya akan cepat lulus. $Q \rightarrow R$

$\therefore$ Jika saya pintar, maka saya akan cepat lulus. $P \rightarrow R$

(B) TABEL KEBENARAN

Diketahui :

Premis 1 : $(P \rightarrow Q)$

Premis 2 : $(Q \rightarrow R)$

Kesimpulan : $(P \rightarrow R)$

No.	P	Q	R	$P \rightarrow Q$	$Q \rightarrow R$	$P \rightarrow R$	$((P \rightarrow Q) \wedge (Q \rightarrow R))$	$((P \rightarrow Q) \wedge (Q \rightarrow R)) \rightarrow (P \rightarrow R)$
1.	T	T	T	T	T	T	T	T
2.	T	T	F	T	F	F	F	T
3.	T	F	T	F	T	T	F	T
4.	T	F	F	F	T	F	F	T
5.	F	T	T	T	T	T	T	T
6.	F	T	F	T	F	T	F	T
7.	F	F	T	T	T	T	T	T
8.	F	F	F	T	T	T	T	T

Baris Kritis : Baris ke 1 karena semua premis/hipotesa bernilai True (T).

Argumen : Jika saya pintar, maka nilai saya akan bagus. Jika nilai saya bagus, maka saya akan cepat lulus.

Argumen valid, karena semua kesimpulan bernilai True (T) dalam baris kritis.